

# PROFESSOR DI WU

✉ di.wu@ucf.edu  
🏠 www.unarylab.com  
📄 <https://scholar.google.com/citations?user=v6DNkTAAAAAJ>

## APPOINTMENT

|                                                                                                          |                   |
|----------------------------------------------------------------------------------------------------------|-------------------|
| <b>Assistant Professor</b>   <i>Electrical and Computer Engineering</i><br>University of Central Florida | 08/2023 – Present |
| <b>Joint Assistant Professor</b>   <i>Computer Science</i><br>University of Central Florida              | 08/2023 – Present |

## RESEARCH INTEREST

- Computer architecture  
— performance, efficiency, etc.
- Emerging computing  
— unary, neuromorphic, quantum, etc.
- Domain specific acceleration  
— artificial intelligence, brain computer interface, etc.
- Heterogeneous system  
— Machine learning systems, multi-GPU systems, etc.

## EDUCATION

|                                                                                                                                                                                                                                                                                                                                                    |                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Doctor of Philosophy</b>   <i>Electrical and Computer Engineering</i><br>University of Wisconsin–Madison <ul style="list-style-type: none"><li>• Advisor: Prof. Joshua San Miguel</li><li>• Thesis: Power-Efficient Computer Architecture via Unary and Approximate Computing</li></ul> 🏆 <b>Harold Peterson Outstanding Dissertation Award</b> | 09/2017 – 07/2023 |
| <b>Master of Engineering</b>   <i>Integrated Circuit Engineering</i><br>Fudan University                                                                                                                                                                                                                                                           | 09/2012 – 01/2015 |
| <b>Bachelor of Science</b>   <i>Microelectronics</i><br>Fudan University                                                                                                                                                                                                                                                                           | 09/2007 – 07/2012 |

## EMPLOYMENT

|                                                                                            |                                      |
|--------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Research Assistant</b><br>Department of Electrical and Computer Engineering, UW–Madison | 09/2017 – 07/2023                    |
| <b>Research Intern</b><br>Cerebras Systems                                                 | 05/2022 – 09/2022, 05/2020 – 09/2020 |
| <b>Research Intern</b><br>Meta (Formerly Facebook)                                         | 05/2019 – 09/2019                    |
| <b>Digital Circuit Engineer</b><br>HiSilicon                                               | 03/2015 – 05/2017                    |
| <b>Research Assistant</b><br>State Key Laboratory of ASIC and System, Fudan University     | 09/2012 – 01/2015                    |

## HONORS AND AWARDS

|                                                                            |      |
|----------------------------------------------------------------------------|------|
| <b>Amar Mukherjee Best Paper Award of ISVLSI</b> 🔗                         | 2025 |
| <b>IEEE Micro Top Pick Honorable Mention</b> 🔗                             | 2025 |
| <b>AMD Faculty Award</b>                                                   | 2024 |
| <b>Harold Peterson Outstanding Dissertation Award at UW–Madison</b> 🔗      | 2024 |
| <b>Distinguished Artifact Evaluation Award of ASPLOS</b> 🔗                 | 2024 |
| <b>MLCommons Machine Learning and Systems Rising Star</b> 🔗                | 2023 |
| <b>Capstone PhD Teaching Award Nomination at UW–Madison</b>                | 2022 |
| <b>Grainger Wisconsin Distinguished Graduate Fellowship at UW–Madison</b>  | 2022 |
| <b>Ph.D. Forum of DAC</b>                                                  | 2021 |
| <b>IEEE Micro Top Pick</b> 🔗                                               | 2021 |
| <b>Gerald Holdridge Outstanding Teaching Assistant Award at UW–Madison</b> | 2020 |
| <b>Chancellor’s Opportunity Fellowship at UW–Madison</b>                   | 2019 |
| <b>Qualcomm Innovation Fellowship Finalist</b> 🔗                           | 2019 |
| <b>Foxconn SmartCity Competition Winner</b>                                | 2019 |
| <b>Hiran Mayukh Award at UW–Madison</b> 🔗                                  | 2018 |
| <b>Rising Star Award at HiSilicon</b>                                      | 2015 |
| <b>National Scholarship at Fudan University (1/67)</b>                     | 2015 |
| <b>Excellent Student Union Leader at Fudan University</b>                  | 2010 |
| <b>Third Prize Freshman Scholarship at Fudan University (3/45)</b>         | 2007 |

+ – Student   \* – Collaborator   × – Advisor   💎 – Top tier   🏆 – Award

## Conference

- [1] Lit Silicon: A Case Where Thermal Imbalance Couples Concurrent Execution in Multiple GPUs [acceptance rate: 19.1%]  
Marco Kurzynski<sup>+</sup>, Shaizeen Aga<sup>\*</sup>, **Di Wu**  
💎 *International Symposium on Computer Architecture*, 2026  
[Open-source software: Lit Silicon](#) [🔗](#)
- [2] CryptOracle: A Modular Framework to Characterize FHE  
Cory Brynds<sup>+</sup>, Parker McLeod<sup>+</sup>, Lauren Caccamise<sup>+</sup>, Asmita Pal, Dewan Saiham, Sazadur Rahman<sup>\*</sup>, Joshua San Miguel<sup>×</sup>, **Di Wu**  
*International Symposium on Performance Analysis of Systems and Software*, 2026  
[Open-source software: CryptOracle](#) [🔗](#)
- [3] Mugi: Value Level Parallelism For Efficient LLMs [acceptance rate: 14.5%]  
Daniel Price<sup>+</sup>, Prabhu Vellaisamy, John Shen<sup>\*</sup>, **Di Wu**  
💎 *International Conference on Architectural Support for Programming Languages and Operating Systems*, 2026, DOI: 10.1145/3779212.3790189  
[Open-source software: artifact](#) [🔗](#)
- [4] PIM-SUM: Fast and Reliable In-Memory Summation for Recommendation Systems  
Fan Li, Ruizhi Zhu, Huize Li, **Di Wu**, Xin Xin<sup>\*</sup>  
*International Conference on Computer Design*, 2025, DOI: 10.1109/ICCD65941.2025.00042
- [5] Can Photonic Interconnects be used for High-Throughput Memory Access in FHE Accelerators?  
Dewan Saiham, Mariam Rabad, **Di Wu**, Sazadur Rahman<sup>\*</sup>  
*International Symposium on Low Power Electronics and Design*, 2025, DOI: 10.1109/ISLPED65674.2025.11261747
- [6] Catwalk: Unary Top-K for Efficient Ramp-No-Leak Neuron Design for Temporal Neural Networks  
Devon Lister<sup>+</sup>, Prabhu Vellaisamy, John Shen<sup>\*</sup>, **Di Wu**  
*IEEE Computer Society Annual Symposium on VLSI*, 2025, DOI: 10.1109/ISVLSI65124.2025.11130314  
🏆 **Amar Mukherjee Best Paper Award**
- [7] Leveraging Photonic Interconnects for Scalable and Efficient Fully Homomorphic Encryption  
Dewan Saiham, **Di Wu**, Sazadur Rahman<sup>\*</sup>  
*Government Microcircuit Applications & Critical Technology Conference*, 2025
- [8] LoAS: Fully Temporal-Parallel Datatflow for Dual-Sparse Spiking Neural Networks [acceptance rate: 22.7%]  
Ruokai Yin, Youngeun Kim, **Di Wu**, Priyadarshini Panda<sup>\*</sup>  
💎 *International Symposium on Microarchitecture*, 2024, DOI: 10.1109/MICRO61859.2024.00084  
[Open-source software: artifact](#) [🔗](#)
- [9] Exploration of Unary Arithmetic-Based Matrix Multiply Units for Low Precision DL Accelerators  
Prabhu Vellaisamy, Harideep Nair, **Di Wu**, Shawn Blanton<sup>\*</sup>, John Paul Shen<sup>\*</sup>  
*IEEE Computer Society Annual Symposium on VLSI*, 2024, DOI: 10.1109/ISVLSI61997.2024.00126
- [10] ALISA: Accelerating Large Language Model Inference via Sparsity-Aware KV Caching [acceptance rate: 19.6%]  
Youpeng Zhao, **Di Wu**, Jun Wang<sup>\*</sup>  
💎 *International Symposium on Computer Architecture*, 2024, DOI: 10.1109/ISCA59077.2024.00077
- [11] Carat: Unlocking Value-Level Parallelism for Multiplier-Free GEMMs [acceptance rate: 20.9%]  
Zhewen Pan, Joshua San Miguel<sup>×</sup>, **Di Wu**  
💎 *International Conference on Architectural Support for Programming Languages and Operating Systems*, 2024, DOI: 10.1145/3620665.3640364  
🏆 **IEEE Micro Top Pick Honorable Mention (24 from all computer architecture papers)**  
🏆 **Distinguished Artifact Evaluation Award**  
[Open-source software: artifact](#) [🔗](#)
- [12] uBrain: A Unary Brain Computer Interface [acceptance rate: 16.6%]  
**Di Wu**, Jingjie Li, Zhewen Pan, Younghyun Kim<sup>\*</sup>, Joshua San Miguel<sup>×</sup>  
💎 *International Symposium on Computer Architecture*, 2022, DOI: 10.1145/3470496.3527401
- [13] uSystolic: Byte-Crawling Unary Systolic Array [acceptance rate: 30.5%]  
**Di Wu**, Joshua San Miguel<sup>×</sup>  
💎 *International Symposium on High-Performance Computer Architecture*, 2022, DOI: 10.1109/HPCA53966.2022.00010  
[Open-source software: uSystolic-Sim](#) [🔗](#)

- [14] When Dataflows Converge: Reconfigurable and Approximate Computing for Emerging Neural Networks  
Di Wu, Joshua San Miguel<sup>×</sup>  
*International Conference on Computer Design*, 2021, DOI: 10.1109/ICCD53106.2021.00014
- [15] UNO: Virtualizing and Unifying Nonlinear Operations for Emerging Neural Networks  
Di Wu, Jingjie Li, Setareh Behrooz, Younghyun Kim\*, Joshua San Miguel<sup>×</sup>  
*International Symposium on Low Power Electronics and Design*, 2021, DOI: 10.1109/ISLPED52811.2021.9502473
- [16] Normalized Stability: A Cross-Level Design Metric for Early Termination in Stochastic Computing  
Di Wu, Ruokai Yin, Joshua San Miguel<sup>×</sup>  
*Asia and South Pacific Design Automation Conference*, 2021, DOI: 10.1145/3394885.3431549
- [17] uGEMM: Unary Computing Architecture for GEMM Applications [acceptance rate: 18.0%]  
Di Wu, Jingjie Li, Ruokai Yin, Hsuan Hsiao, Younghyun Kim\*, Joshua San Miguel<sup>×</sup>  
♦ *International Symposium on Computer Architecture*, 2020, DOI: 10.1109/ISCA45697.2020.00040  
✂ IEEE Micro Top Pick (12 from all computer architecture papers)  
Open-source software: UnarySim [↗](#)
- [18] Approximate Hardware Techniques for Energy-Quality Scaling Across the System  
Younghyun Kim\*, Joshua San Miguel<sup>×</sup>, Setareh Behrooz, Tianen Chen, Kyuin Lee, Yongwoo Lee, Jingjie Li, Di Wu  
*International Conference on Electronics, Information, and Communication*, 2020, DOI: 10.1109/ICEIC49074.2020.9051208
- [19] SECO: A Scalable Accuracy Approximate Exponential Function Via Cross-Layer Optimization  
Di Wu, Tianen Chen, Chienfu Chen, Oghenefego Ahia, Joshua San Miguel<sup>×</sup>, Mikko Lipasti\*, Younghyun Kim\*  
*International Symposium on Low Power Electronics and Design*, 2019, DOI: 10.1109/ISLPED.2019.8824959
- [20] In-Stream Stochastic Division and Square Root via Correlation [acceptance rate: 18.9%]  
Di Wu, Joshua San Miguel<sup>×</sup>  
*Design Automation Conference*, 2019, DOI: 10.1145/3316781.3317844
- [21] Convergence-Optimized Variable Node Structure for Stochastic LDPC Decoder  
Qichen Zhang, Yun Chen, Di Wu, Xiaoyang Zeng, Yeong-luh Ueng  
*International Conference on Acoustics, Speech and Signal Processing*, 2016, DOI: 10.1109/ICASSP.2016.7472936
- [22] An Area-Efficient Architecture for Stochastic LDPC Decoder  
Qichen Zhang, Yun Chen, Di Wu, Xiaoyang Zeng, Yeong-luh Ueng  
*International Conference on Digital Signal Processing*, 2015, DOI: 10.1109/ICDSP.2015.7251868
- [23] Latency-Optimized Stochastic LDPC Decoder for High-Throughput Applications  
Di Wu, Yun Chen, Qichen Zhang, Lirong Zheng, Xiaoyang Zeng, Yeong-luh Ueng  
*International Symposium on Circuits and Systems*, 2015, DOI: 10.1109/ISCAS.2015.7169329
- [24] A High-Throughput LDPC Decoder for Optical Communication  
Di Wu, Yun Chen, Yuebin Huang, Yeongluh Ueng, Lirong Zheng, Xiaoyang Zeng  
*International Conference on ASIC*, 2013, DOI: 10.1109/ASICON.2013.6811973

## Journal

- [1] OptoLink: Breaking Memory Bandwidth Bottlenecks in FHE Accelerators with Photonic Interconnects  
Dewan Saiham, Di Wu, Sazadur Rahman\*  
*ACM Transactions on Design Automation of Electronic Systems* (2026), DOI: <https://doi.org/10.1145/3806395>
- [2] Towards Plug & Play Myoelectric Control via One-Shot Latent Representation Alignment  
Zubaidah Al-Mashhadani<sup>+</sup>, Trevor Overton, Di Wu, Mohsen Rakhshan\*  
♦ *IEEE Transactions on Neural Systems & Rehabilitation Engineering* 33 (2025), pp. 4409–4421, DOI: 10.1109/TNSRE.2025.3626255
- [3] Synergizing Quantum Techniques with Machine Learning for Advancing Drug Discovery Challenge  
Zhiding Liang, Zichang He, Yue Sun, Dylan Herman, Qingyue Jiao, Yanzhang Zhu<sup>+</sup>, Weiwen Jiang\*, Xiaowei Xu\*, Di Wu, Marco Pistoia\*, Yiyu Shi\*  
*Scientific Reports* 14.1 (2024), p. 31216, DOI: 10.1038/s41598-024-82576-4
- [4] uGEMM: Unary Computing for GEMM Applications  
Di Wu, Jingjie Li, Ruokai Yin, Hsuan Hsiao, Younghyun Kim\*, Joshua San Miguel<sup>×</sup>  
*IEEE Micro* 41.3 (2021), pp. 50–56, DOI: 10.1109/MM.2021.3065369  
✂ IEEE Micro Top Pick (12 from all computer architecture papers)
- [5] In-Stream Correlation-Based Division and Bit-Inserting Square Root in Stochastic Computing  
Di Wu, Ruokai Yin, Joshua San Miguel<sup>×</sup>  
*IEEE Design & Test* 38.6 (2021), pp. 53–59, DOI: 10.1109/MDAT.2021.3050716
- [6] Strategies for Reducing Decoding Cycles in Stochastic LDPC Decoders  
Di Wu, Yun Chen, Qichen Zhang, Yeong-luh Ueng, Xiaoyang Zeng  
*IEEE Transactions on Circuits and Systems II: Express Briefs* 63.9 (2016), pp. 873–877, DOI: 10.1109/TCSII.2016.2535038

- [7] An Efficient Multirate LDPC-CC Decoder With a Layered Decoding Algorithm for the IEEE 1901 Standard  
Yun Chen, Qichen Zhang, **Di Wu**, Changsheng Zhou, Xiaoyang Zeng  
*IEEE Transactions on Circuits and Systems II: Express Briefs* 61.12 (2014), pp. 992–996, DOI: 10.1109/TCSII.2014.2362721

## Workshop and Poster

- [1] PyTorch-Based Implementation of Relay-BP  
Thomas Danielsen<sup>+</sup>, Caden Brock<sup>+</sup>, Yanzhang Zhu<sup>+</sup>, **Di Wu**  
*Florida Quantum Conference* (2026)
- [2] Mugi: Value Level Parallelism For Nonlinear Operations in LLMs  
Daniel Price<sup>+</sup>, Prabhu Vellaisamy, John Shen<sup>\*</sup>, **Di Wu**  
*Workshop on Unary Computing, collocated with ASPLOS* (2026)
- [3] Agraph: A unified Graph Representation for At-Will Simulation of Emerging Stacks  
Daniel Price<sup>+</sup>, Prabhu Vellaisamy, Patricia Gonzalez<sup>\*</sup>, George Michelogiannakis<sup>\*</sup>, John Shen<sup>\*</sup>, **Di Wu**  
*Workshop on Unary Computing, collocated with ASPLOS* (2026)
- [4] Sense-as-You-Go: A Neuromorphic Framework for Efficient Edge Sensing and Processing  
**Di Wu**, Mohsen Rakhshan<sup>\*</sup>  
*Energy Consequences of Information Workshop* (2026)
- [5] A-Graph: A Unified System Representation for Automated Cross-Stack Optimization  
**Di Wu**  
*Energy Consequences of Information Workshop* (2026)
- [6] Catwalk Neuron for Efficient Temporal Neural Networks  
**Di Wu**  
*Energy Consequences of Information Workshop* (2026)
- [7] EnerQy: Energy Estimation for Quantum Computing  
**Di Wu**, Siyuan Niu<sup>\*</sup>, Zhipeng Deng<sup>\*</sup>  
*Energy Consequences of Information Workshop* (2026)
- [8] Syndrilla: Simulating Decoders for Quantum Error Correction using PyTorch  
Yanzhang Zhu<sup>+</sup>, Chen-Yu Peng, Yun Hao Chen, Siyuan Niu<sup>\*</sup>, Yeong-Luh Ueng<sup>\*</sup>, **Di Wu**  
*IEEE International Conference on Quantum Computing and Engineering*, 2025, DOI: 10.1109/QCE65121.2025.10470  
[Open-source software: Syndrilla](#) 
- [9] Are We Scaling the Right Thing? A System Perspective on Test-Time Scaling  
Youpeng Zhao, Jinpeng LV<sup>\*</sup>, **Di Wu**, Jun Wang<sup>\*</sup>, Christopher Gooley<sup>\*</sup>  
*NeurIPS Workshop on Efficient Reasoning* (2025)
- [10] Synergizing Error Suppression, Mitigation and Correction for Fault-Tolerant Quantum Computing  
Yanzhang Zhu<sup>+</sup>, Siyuan Niu<sup>\*</sup>, **Di Wu**  
*Workshop on Quantum Intelligence, Learning & Security, collocated with IEEE TPS* (2024), DOI: 10.1109/TPS-ISA62245.2024.00065
- [11] Exploration of Unary Arithmetic-Based Matrix Multiply Units for Low Precision DL Accelerators  
Prabhu Vellaisamy, Harideep Nair, **Di Wu**, Shawn Blanton<sup>\*</sup>, John Paul Shen<sup>\*</sup>  
*Workshop on Unary Computing, collocated with ASPLOS* (2024)
- [12] xBrain: Brain-Like Computing for Explainable Brain-Computer Interfaces  
Queenly Xie<sup>+</sup>, Prabhu Vellaisamy, **Di Wu**  
*Young Architect Workshop, collocated with ASPLOS* (2024)
- [13] T-MAC: Temporal Multiplication with Accumulation  
Zhewen Pan, **Di Wu**, Joshua San Miguel<sup>×</sup>  
*Young Architect Workshop, collocated with ASPLOS* (2022)

## Pre-Print

- [1] Lottery BP: Unlocking Quantum Error Decoding at Scale  
Yanzhang Zhu<sup>+</sup>, Chen-Yu Peng, Yun Hao Chen, Yeong-Luh Ueng<sup>\*</sup>, **Di Wu**  
*arXiv Pre-print* (2026)  
[Open-source software: Syndrilla](#) 
- [2] uSense: Unary-Computing-based Stochastic Edge Neuromorphic Sensing  
Zubaidah Al-Mashhadani<sup>+</sup>, Mohsen Rakhshan<sup>\*</sup>, **Di Wu**  
*Research Square Pre-print* (2026)
- [3] A-Graph: A Unified Graph Representation for At-Will Simulation across System Stacks  
Daniel Price<sup>+</sup>, Prabhu Vellaisamy, Patricia Gonzalez<sup>\*</sup>, George Michelogiannakis<sup>\*</sup>, John Shen<sup>\*</sup>, **Di Wu**  
*arXiv Pre-print* (2026)  
[Open-source software: Archx](#) 

- [4] Chopper: A Multi-Level GPU Characterization Tool & Derived Insights Into LLM Training Inefficiency  
Marco Kurzynski<sup>+</sup>, Shaizeen Aga<sup>\*</sup>, **Di Wu**  
*arXiv Pre-print* (2025)  
[Open-source software: Chopper](#)
- [5] Unleashing The Potential of LLMs for Quantum Computing: A Study in Quantum Architecture Design  
Zhiding Liang, Jinglei Cheng, Rui Yang, Hang Ren, Zhixin Song, **Di Wu**, Tongyang Li<sup>\*</sup>, Yiyu Shi<sup>\*</sup>  
*arXiv Pre-print* (2023)
- [6] Representation Range Needs for 16-Bit Neural Network Training  
Valentina Popescu<sup>\*</sup>, Abhinav Venigalla<sup>\*</sup>, **Di Wu**, Robert Schreiber<sup>\*</sup>  
*arXiv Pre-print* (2021)  
[Open-source software: production-level cbfloat16](#)

## Patent

- [1] Computer Architecture with Value-Level Parallelism  
Joshua San Miguel, Di Wu, Zhewen Pan  
US Patent Application 18/221,127, 2025

## INVITED TALKS

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### Lit Silicon: How Thermal Variation Impairs AI Performance via Concurrent Execution in GPU Nodes and Its Simple Repairs

AMD Research and Advanced Development 05/2026

### Value-Level Parallelism: New Opportunities for Parallel Computing from Fewer Bits in the AI Era

University of Pittsburgh 03/2026

University of Wisconsin–Madison 12/2025

Lawrence Berkeley National Laboratory 10/2025

NVIDIA Research 10/2025

### The Phantom of the Datacenter: Unmasking the Culprit behind Performance Variation in GenAI Training

AMD Research and Advanced Development 10/2025

### Salvage Deep Learning Efficiency: A Unary Computing Approach

University of California, Santa Cruz 02/2025

ShanghaiTech University 12/2024

Fudan University 12/2024

Case Western Reserve University 10/2024

Peking University 08/2024

University of Minnesota Twin Cities 03/2024

University of Louisiana at Lafayette 11/2023

### Unary Computing for Power-Efficient Computer Architecture

AMD Research and Advanced Development 07/2023

University of Central Florida 02/2023

University of California, Los Angeles 11/2022

University of California, Santa Barbara 10/2022

## TEACHING AND MENTORING

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### Instructor

EEL5796 (Big Data Computer Architecture and Systems), UCF SP 2026

EEL4768 (Computer Architecture), UCF SP 2026

EEE3342C (Digital Systems), UCF FA 2025

EEL4742C (Embedded Systems), UCF SP 2025

EEE3342C (Digital Systems), UCF FA 2024

EEL5796 (Big Data Computer Architecture and Systems), UCF SP 2024

ECE697 (Capstone Project in Machine Learning and Signal Processing), UW–Madison SU 2023

### Teaching Assistant

ECE554 (Digital Engineering Lab), UW–Madison SP 2022

ECE454 (Mobile Computing Lab), UW–Madison FA 2021

ECE454 (Mobile Computing Lab), UW–Madison FA 2020

ECE554 (Digital Engineering Lab), UW–Madison SP 2020

ECE554 (Digital Engineering Lab), UW–Madison FA 2019

ECE554 (Digital Engineering Lab), UW–Madison SP 2019

ECE554 (Digital Engineering Lab), UW–Madison FA 2018

ECE552 (Introduction to Computer Architecture), UW–Madison FA 2018

### Guest Lecturer

ECE18743 (Neuromorphic Computer Architecture & Processor Design), CMU SP 2025

ECE757 (Advanced Computer Architecture II), UW–Madison SP 2021

ECE757 (Advanced Computer Architecture II), UW–Madison SP 2020

ECE752 (Advanced Computer Architecture I), UW–Madison SP 2019



## PROFESSIONAL SERVICE

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### Panelist

|                                                                     |      |
|---------------------------------------------------------------------|------|
| NSF Medium Panel in Division of Computer and Network Systems (CNS)  | 2024 |
| IEEE Workshop on Quantum Intelligence, Learning & Security (QUILLS) | 2024 |

### Committee

|                                                                                                 |                        |
|-------------------------------------------------------------------------------------------------|------------------------|
| Organizing Chair of Artifact Evaluation at ISPASS                                               | 2026                   |
| Organizing Chair of Workshop on Architecting Error Corrected Quantum Computers (ARQTEC) at HPCA | 2025                   |
| Organizing Chair of Undergrad Panel on "Charging STEM Career" at UCF                            | 2024                   |
| Organizing Chair of Workshop on Unary Computing (WUC) at ASPLOS                                 | 2024, 2026             |
| Organizing Committee of Quantum Computing for Drug Discovery Challenge at ICCAD                 | 2023                   |
| Program Committee of ASPLOS                                                                     | 2026                   |
| Program Committee of HPCA                                                                       | 2024, 2025, 2026       |
| Program Committee of ISCA                                                                       | 2024, 2025, 2026       |
| Program Committee of MICRO                                                                      | 2025, 2026             |
| Program Committee of ICS                                                                        | 2026                   |
| Program Committee of IISWC                                                                      | 2023, 2025             |
| Program Committee of ISPASS                                                                     | 2024, 2025, 2026       |
| Program Committee of IPDPS                                                                      | 2025                   |
| Program Committee of DAC                                                                        | 2025                   |
| Program Committee of ICCAD                                                                      | 2024                   |
| Program Committee of DAC PhD Forum                                                              | 2024                   |
| Program Committee of ICCD                                                                       | 2023, 2024             |
| Program Committee of GLSVLSI                                                                    | 2026                   |
| Program Committee of ICRC                                                                       | 2024                   |
| Program Committee of ICA3PP                                                                     | 2023                   |
| Program Committee of IEEE Workshop on Quantum Intelligence, Learning & Security (QUILLS)        | 2024                   |
| Program Committee of Young Architect Workshop (YArch)                                           | 2023, 2024, 2025, 2026 |
| Program Committee of ASPLOS Artifact Evaluation                                                 | 2020, 2021             |
| Program Committee of MICRO Artifact Evaluation                                                  | 2021                   |

### Mentor

|                                                                                    |            |
|------------------------------------------------------------------------------------|------------|
| UCF IEEE Engineering in Medicine and Biology Society (EMBS) Student Branch Chapter | 2024       |
| Computer Architecture Long-term Mentoring (CALM)                                   | 2024       |
| Young Architect Workshop (YArch)                                                   | 2023       |
| Undergrad Architecture Mentoring Workshop (uArch)                                  | 2023, 2024 |
| MICRO – "Meet a Senior PhD Student"                                                | 2020       |

### Journal Reviewer

|                                                                                      |
|--------------------------------------------------------------------------------------|
| ACM Transactions on Architecture and Code Optimization (TACO)                        |
| ACM Transactions on Embedded Computing Systems (TECS)                                |
| ACM Transactions on Reconfigurable Technology and Systems (TRETs)                    |
| IEEE Computer Architecture Letter (CAL)                                              |
| IEEE Transactions on Circuits and Systems I (TCAS-I)                                 |
| IEEE Transactions on Computers (TC)                                                  |
| IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD) |
| IEEE Transactions on Emerging Topics in Computing (TETC)                             |
| IEEE Transactions on Very Large Scale Integration (VLSI) Systems (TVLSI)             |
| IEEE Signal Processing Letters (SPL)                                                 |
| Journal of Network and Computer Applications (JNCA)                                  |
| npj Quantum Information                                                              |
| npj Unconventional Computing                                                         |

## FUNDING

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|                                                                                                     |                  |
|-----------------------------------------------------------------------------------------------------|------------------|
| <b>DOE: APT-Q: Algorithm-Architecture Co-design in Partial Fault-Tolerance for Quantum Utility</b>  | 2026             |
| Co-PI: \$125k/\$400k                                                                                |                  |
| <b>NSF EAGER: SENSE: National Security Evaluation of Neurotechnology Systems and Emerging Tools</b> | 2025             |
| Co-PI: \$100k/\$300k                                                                                |                  |
| <b>AMD Faculty Award (unrestricted gift)</b>                                                        | 2024             |
| PI: \$100k                                                                                          |                  |
| <b>AMD AI &amp; HPC Cluster Award</b>                                                               | 2024, 2025, 2026 |
| <b>AMD Developer Cloud Credit: \$2k</b>                                                             | 2025             |
| <b>Quantum Computing Access at NERSC (QCAN) Program</b>                                             | 2025             |
| <b>NSF ACCESS Program</b>                                                                           | 2024             |

## STUDENTS

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### Current PhD Students

|                        |                |
|------------------------|----------------|
| Yang Yu                | 2026 – Present |
| Chetan Choudhary       | 2025 – Present |
| Daniel Price           | 2024 – Present |
| Marco Kurzynski        | 2024 – Present |
| Yanzhang Zhu           | 2024 – Present |
| Zubaidah Al-Mashhadani | 2024 – Present |

### Former Students

|                                             |             |
|---------------------------------------------|-------------|
| Cory Brynds (BS/MS@UCF → full-time@AMD)     | 2023 – 2026 |
| Lauren Caccamise (BS@UCF → PhD@Purdue)      | 2024 – 2025 |
| Parker McLeod (BS@UCF → full-time@AMD)      | 2023 – 2024 |
| Tyler Goldsmith (BS@UCF → full-time@AMD)    | 2023 – 2024 |
| Mustafa Nisar (BS@UoT → co-op@AMD)          | 2023 – 2024 |
| Zhewen Pan (MS@UW–Madison → PhD@UW–Madison) | 2020 – 2022 |
| Ruokai Yin (BS@UW–Madison → PhD@Yale)       | 2019 – 2021 |